

Jyotirmay Nag Setu

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Research Interests

Augmented Reality (AR), Virtual Reality (VR), Extended Reality (XR), Deep Learning, Computer Vision, Human-Computer Interaction, Multimodal Sensing, Intelligent Extended Reality

Education

University of Texas at San Antonio *San Antonio, TX*
Ph.D. Candidate, Computer Science — GPA: 4.00/4.00 Jan 2023 – Dec 2027 (Expected)

- Advisor: Dr. John Quarles, SAVE Lab
- Dissertation focus: Cybersickness prediction, cognitive load estimation, and affective modeling in VR using multimodal physiological signals and deep learning

Technical Skills

ML / DL	PyTorch, TensorFlow, Keras; CNNs, LSTMs, GRUs, Transformers, Transfer Learning, Temporal Modeling, Multi-task Learning
Computer Vision	OpenCV, feature extraction, video analysis, scene understanding, image classification
AR / VR / XR	Unity 3D, Meta Quest, HTC Vive, HMD data collection, gaze tracking, physiological sensors (EEG, GSR, HR)
NLP / LLM	Time series LLMs (TimeGPT, Chronos), fine-tuning, SHAP-based explainability
Languages	Python, Kotlin, Java, Shell scripting, PL/SQL
Data / Cloud	PostgreSQL, Oracle DB, Hadoop, FastAPI, Docker, ETL pipelines, SAP BO
Tools	Git, LaTeX, Jupyter, Android (Jetpack Compose)

Research Experience

Graduate Research Assistant - SAVE Lab, UTSA *Jan 2023 – Present*
Supervisor: Dr. John Quarles

- Designed and deployed end-to-end pipelines for multimodal VR data collection capturing head motion, eye tracking, EEG, GSR, and heart rate signals to study cybersickness and cognitive load.
- Developed deep learning models (LSTM, GRU, MLP, Transformer) for real-time cybersickness severity classification and cognitive load prediction from physiological time-series data.
- Fine-tuned state-of-the-art time-series foundation models (TimeGPT, Amazon Chronos) on proprietary VR datasets for affective state forecasting.
- Applied SHAP-based feature importance analysis to interpret model decisions and improve clinical transparency of VR health assessments.
- Leveraged computer vision to analyze VR scene characteristics (optical flow, scene complexity, depth cues) and identify visual factors contributing to cybersickness.
- Co-developed *PatchFusionVR*: a multi-task deep learning framework predicting gaze, reaction time, and cognitive load simultaneously from multimodal VR signals (VRST 2025).
- Collected and released *Mazed and Confused*: a large-scale VR dataset of cybersickness, working memory, and attention during real-walking tasks (IEEE ISMAR 2024).
- Developed *ForeSightVR*, a forecasting system predicting cybersickness and cognitive overload in VR multitasking environments (IEEE AIXR 2026).
- Proposed a statistical abstraction framework for integrating heterogeneous VR datasets in ordinal cybersickness prediction to enable cross-study generalization (IEEE VR 2026, pp. 668–677).

Industry Experience

Data Engineer — Bangladesh Bank, R&D Data Science Applications Wing

Nov 2020 – Ongoing

- Architected and maintained fully automated Python data pipelines processing millions of financial records daily.
- Built Oracle Data Integrator (ODI) and Oracle Warehouse Builder (OWB) mappings for large-scale ETL workflows.
- Developed Oracle BI dashboards for regulatory reporting and analytics.

Data Engineer — Robi Axiata Ltd., EDW & Analytics Team

May 2020 – Nov 2020

- Designed ETL processes and wrote Python/Shell scripts for high-volume telecom data processing.
- Processed big data using Hadoop and generated business intelligence reports via SAP BO.
- Maintained PL/SQL procedures in Oracle Database for data transformation.

Publications

2026

- [1] **Setu, J. N.**, Islam, R., & Quarles, J. “ForeSightVR: Forecasting Cybersickness and Cognitive Overload in VR Multitasking.” *IEEE International Conference on Artificial Intelligence and eXtended Reality (AIxVR)*, 2026.
- [2] **Setu, J. N.**, & Quarles, J. “A Statistical Abstraction Framework for Integrating Heterogeneous VR Datasets in Ordinal Cybersickness Prediction.” *IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*, pp. 668–677, 2026.
- [3] Saha, P., **Setu, J. N.**, Dev, A. K., Islam, R., Tabar, M., & Quarles, J. “A Comparative Study of Data Augmentation Methods for Improving Cybersickness Prediction.” *IEEE International Conference on Artificial Intelligence and eXtended Reality (AIxVR)*, 2026.

2025

- [1] **Setu, J. N.**, Le, J. M., Kundu, R. K., Giesbrecht, B., Höllerer, T., Hoque, K. A., ... Quarles, J. “Predicting and Explaining Cognitive Load, Attention, and Working Memory in Virtual Multitasking.” *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 5, pp. 3014–3024, 2025.
- [2] Pavel, M. I., Mahmud, M. R., **Setu, J. N.**, Desai, K., & Quarles, J. “PatchFusionVR: Multitask Prediction of User Gaze, Reaction Time, and Cognitive Load in Virtual Reality from Multimodal Signals.” *ACM Symposium on Virtual Reality Software and Technology (VRST)*, 2025.
- [3] Dev, A. K., Saha, P., **Setu, J. N.**, Ang, S., Quarles, J., & Tabar, M. “Investigating the Potential of Multi-task Learning for Enhancing SSQ Prediction.” *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 815–823, 2025.
- [4] **Setu, J. N.**, Desai, K., & Quarles, J. “Towards Cybersickness Severity Classification from VR Gameplay Videos Using Transfer Learning and Temporal Modeling.” arXiv:2510.10422. *IEEE ISMAR TRUST-XR Workshop*, 2025.

2024

- [1] **Setu, J. N.**, Le, J. M., Kundu, R. K., Giesbrecht, B., Höllerer, T., Hoque, K. A., ... Quarles, J. “Mazed and Confused: A Dataset of Cybersickness, Working Memory, Mental Load, Physical Load, and Attention During a Real Walking Task in VR.” *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 1048–1057, 2024.

2019

- [1] Islam, M. M., Debnath, A., Al Sayeed, T., **Setu, J. N.**, et al. “A Gray Box Interpretable Debugging Approach for Deep Sequence Learning Model.” *ICLR Workshop: Debugging Machine Learning Models (Poster)*, 2019.

Undergraduate Thesis

Sakib, M. S., **Setu, J. N.**, Islam, M. M. “Multi-level Balanced Caching Approach to Optimize Repeated Subgraph Queries,” 2018.

Service & Volunteer Work

- Student Volunteer, IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR), 2024 & 2025
- Reviewer, ISMAR 2025 & 2026, VRST 2025, IEEEVR 2024 & 2025, CHI 2025
- Contest Proctor, ICPC Dhaka Regional, 2021

Certifications

- Complete Machine Learning & Data Science with Python | ML A-Z (LinkedIn Learning)
- NLP with Python for Machine Learning Essential Training (LinkedIn Learning)
- OpenCV for Python Developers (LinkedIn Learning)

Language Proficiency

TOEFL iBT (Nov 2021): Total 107/120 Reading: 30/30 Listening: 29/30 Speaking: 25/30 Writing: 23/30